

Location, Time, and Observer Based Three-Valued Logic: A Minimal Formal System Where Truth is Never Absolute

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Abstract

We present **LTO-K3**, a minimal formal system in which the truth value of a proposition is never absolute but is always evaluated with respect to three explicit parameters: a *location*, a *time*, and an *observer*. The system admits exactly three truth values—**T** (true), **F** (false), and **U** (unknown)—composed via Kleene’s strong three-valued connectives. We give a formal syntax, a compositional point-wise semantics $\widehat{V} : \mathcal{L} \times L \times \Theta \times O \rightarrow \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$, two consequence relations (local and universal), and prove a small body of results: information-monotonicity, conservativity over classical logic, locality, absence of **T**-tautologies (a known property of K_3 that we re-derive in the indexed setting), and the De Morgan laws. We are careful to distinguish what is genuinely new (the *packaging*: location and observer as non-optional, concrete parameters; a deliberate refusal to introduce modal operators) from what is inherited from established work (Kleene semantics [2], modal/temporal/epistemic indexing [3, 4, 5], three-valued spatio-temporal logics in verification). The contribution is therefore a careful synthesis, not a discovery, and we name its limitations explicitly.

Keywords: three-valued logic, Kleene logic, indexed semantics, temporal logic, epistemic logic, contextual semantics, partial information.

1 Introduction

Classical two-valued logic forces every proposition into one of two states: *true* or *false*. In many practical settings—verification of distributed systems, knowledge representation, robotics, legal reasoning, data quality—this dichotomy is too coarse. Truth often depends on *where* an evaluation is performed, *when* it is performed, and *who* performs it, and in many cases the evaluator simply does not know.

We introduce a minimal logic, **LTO-K3** (Location, Time, Observer; Kleene three-valued), that takes these dependencies as primitive. Every atomic proposition is evaluated under an explicit tuple (l, θ, o) of location, time, and observer, and may take one of three values: true, false, or unknown. Compound formulas are evaluated point-wise via Kleene’s strong three-valued tables.

The system is intentionally austere. It contains no modal operators, no accessibility relations, no transition systems, and no quantification over locations, times, or observers. Its purpose is to serve as a compact, unambiguous core on which richer domain-specific layers (synchronization, trust, frame alignment, modal operators) may later be built.

Contributions.

1. A formal syntax and compositional point-wise semantics for a three-valued indexed logic over (L, Θ, O) frames (Sections 3–4).
2. Two consequence relations—local at a frame and universal across frames—based on the designated value **T** (Section 5).

3. A small body of meta-theoretic results (Section 6): information-monotonicity, locality, conservativity over classical logic, De Morgan laws, and the absence of **T**-tautologies in the unindexed fragment.
4. An explicit comparison with the closest neighbours: Kleene K_3 , Łukasiewicz L_3 , Belnap’s **FOUR**, Priest’s LP, Bochvar B_3 , hybrid logic, and multi-agent epistemic temporal logic (Section 8).
5. A frank limitations section (Section 9).

What is and is not new. None of the parts is new. Three-valued logics go back to Łukasiewicz [1] and Kleene [2]. Temporal indexing goes back to Prior [3]; possible-worlds indexing to Kripke [5]; agent indexing to Hintikka [4] and, in computer science, to Fagin et al. [11]. What is new is the *packaging*: making location, time, and observer all simultaneously non-optional, concrete, first-class parameters of a single valuation function, and deliberately refusing to lift them with modal operators. We state this honestly throughout.

2 Preliminaries: Kleene’s Strong Three-Valued Logic

Kleene’s strong three-valued logic K_3 [2] extends classical two-valued logic with a third value **U** (“unknown” or “undefined”) and defines the connectives so that **U** propagates conservatively: a connective returns **T** (resp. **F**) iff every way of resolving each **U** to **T** or **F** would yield **T** (resp. **F**); otherwise it returns **U**. This “supervaluation-like” character is what makes K_3 the natural choice when **U** is read epistemically as “not yet known”. The truth tables are:

$\wedge$	T	U	F	$\vee$	T	U	F	p	$\neg p$	$\rightarrow$	T	U	F
T	T	U	F	T	T	T	T	T	F	T	T	U	F
U	U	U	F	U	T	U	U	U	U	U	T	U	U
F	F	F	F	F	T	U	F	F	T	F	T	T	T

Material implication is defined as $\varphi \rightarrow \psi := \neg\varphi \vee \psi$. We use this strong-Kleene implication throughout; for the contrast with Łukasiewicz’s L_3 (where $\mathbf{U} \rightarrow \mathbf{U} = \mathbf{T}$) see Section 8.

Information ordering. Define the partial order $\sqsubseteq$ on $\{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$ by $\mathbf{U} \sqsubseteq \mathbf{T}$, $\mathbf{U} \sqsubseteq \mathbf{F}$, and $x \sqsubseteq x$ for all x ; **T** and **F** are incomparable. Intuitively, $x \sqsubseteq y$ means “ y contains at least as much information as x .” All Kleene connectives are *monotone* in $\sqsubseteq$ component-wise—resolving an **U** to **T** or **F** never reverses an already-determined truth value. This monotonicity is the formal reason K_3 is appropriate when **U** denotes epistemic uncertainty.

3 Syntax

Fix a countably infinite set AP of *atomic propositions* $p, q, r, \dots$. The language $\mathcal{L}$ is generated by the grammar

$$\varphi, \psi ::= p \mid \neg\varphi \mid (\varphi \wedge \psi) \mid (\varphi \vee \psi) \mid (\varphi \rightarrow \psi), \quad p \in \text{AP}.$$

We omit parentheses under the usual precedence $\neg > \wedge > \vee > \rightarrow$. The language has no constants, no biconditional, and no quantifiers. The set of atomic propositions occurring in φ is written $\text{AP}(\varphi)$.

Remark 1. The biconditional $\varphi \leftrightarrow \psi$ and the quantifiers $\forall, \exists$ are deliberately omitted. The biconditional can be defined as $(\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi)$ but its K_3 behaviour is non-standard (it is not the equivalence relation classically expected) and we prefer to omit it from the core.

Quantifiers would force a domain commitment and are orthogonal to the indexing concern of this paper. Crucially, the language contains *no modal operators* indexed over L , Θ , or O (no $\Box_l$, no $F\theta$, no $K_o\varphi$). This is the essential design choice: indices appear in the meta-language (the semantics), not in the object language.

4 Semantics

4.1 Frames and models

Definition 2 (Frame). A *frame* is a triple $\mathcal{F} = (L, \Theta, O)$ where L , Θ , and O are non-empty sets of *locations*, *times*, and *observers*, respectively. We call $(l, \theta, o) \in L \times \Theta \times O$ a *point* of the frame.

We impose no structure on L , Θ , or O . In particular Θ is not required to be ordered, L not required to carry a topology, and O not required to carry an information-state relation. Any such structure is a domain-specific extension; see Section 9.

Definition 3 (Model). An *LTO-K3 model* is a pair $\mathcal{M} = (\mathcal{F}, V)$ where $\mathcal{F} = (L, \Theta, O)$ is a frame and

$$V : \text{AP} \times L \times \Theta \times O \longrightarrow \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$$

is a *base valuation* that assigns a truth value to each atomic proposition at each point. Note that V is total: every atomic proposition has a value (possibly $\mathbf{U}$) at every point.

4.2 Compositional valuation

The base valuation V extends uniquely to a valuation $\widehat{V}$ on all formulas by structural recursion using the Kleene tables of Section 2.

Definition 4 (Extended valuation). Let $\mathcal{M} = (\mathcal{F}, V)$ be a model with $\mathcal{F} = (L, \Theta, O)$. The *extended valuation* $\widehat{V} : \mathcal{L} \times L \times \Theta \times O \rightarrow \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$ is defined by recursion on φ :

$$\begin{aligned} \widehat{V}(p, l, \theta, o) &= V(p, l, \theta, o), \quad p \in \text{AP}, \\ \widehat{V}(\neg\varphi, l, \theta, o) &= \neg_{K_3} \widehat{V}(\varphi, l, \theta, o), \\ \widehat{V}(\varphi \wedge \psi, l, \theta, o) &= \widehat{V}(\varphi, l, \theta, o) \wedge_{K_3} \widehat{V}(\psi, l, \theta, o), \\ \widehat{V}(\varphi \vee \psi, l, \theta, o) &= \widehat{V}(\varphi, l, \theta, o) \vee_{K_3} \widehat{V}(\psi, l, \theta, o), \\ \widehat{V}(\varphi \rightarrow \psi, l, \theta, o) &= \widehat{V}(\varphi, l, \theta, o) \rightarrow_{K_3} \widehat{V}(\psi, l, \theta, o), \end{aligned}$$

where $\neg_{K_3}, \wedge_{K_3}, \vee_{K_3}, \rightarrow_{K_3}$ are the connectives of Section 2.

This is the entire semantics. Crucially, the recursion is *point-wise*: both sides of every binary connective are evaluated at *the same point* (l, θ, o) . There is no syntactic mechanism in $\mathcal{L}$ for referring to a different point.

Remark 5 (Cross-frame composition). A natural question is how to combine $\widehat{V}(\varphi, l_1, \theta_1, o_1)$ with $\widehat{V}(\psi, l_2, \theta_2, o_2)$. The answer in LTO-K3 is: *the language does not express such combinations*. There is no formula whose two sub-formulas are evaluated at different points; the recursion above forces a single (l, θ, o) throughout. Defining cross-frame operators would require explicit bridging rules (synchronization, trust, observer alignment) that are domain-specific. Any such extension is outside the core system.

5 Consequence Relations

We take $\mathbf{T}$ as the sole designated value. (Designating $\{\mathbf{T}, \mathbf{U}\}$ gives Priest's LP [8], a different logic; see Section 8.)

Definition 6 (Local satisfaction). Let $\mathcal{M} = (\mathcal{F}, V)$ be a model and (l, θ, o) a point. We say φ is *satisfied at* (l, θ, o) in $\mathcal{M}$, written $\mathcal{M}, (l, \theta, o) \models \varphi$, iff $\widehat{V}(\varphi, l, \theta, o) = \mathbf{T}$.

Definition 7 (Consequence). Let $\Gamma \subseteq \mathcal{L}$ and $\varphi \in \mathcal{L}$.

- *Local entailment at* (l, θ, o) in $\mathcal{M}$: $\Gamma \models_{\mathcal{M}, (l, \theta, o)} \varphi$ iff $\mathcal{M}, (l, \theta, o) \models \gamma$ for every $\gamma \in \Gamma$ implies $\mathcal{M}, (l, \theta, o) \models \varphi$.
- *Universal entailment*: $\Gamma \models \varphi$ iff $\Gamma \models_{\mathcal{M}, (l, \theta, o)} \varphi$ for every model $\mathcal{M}$ and every point (l, θ, o) of $\mathcal{M}$.

These are the standard K_3 -style consequence relations, lifted point-wise over the frame.

6 Properties

We collect the basic meta-theoretic properties of LTO- K_3 . The proofs are short because the semantics is point-wise: at each fixed (l, θ, o) the system collapses to ordinary K_3 , so every K_3 identity lifts.

Lemma 8 (Locality). *For every model $\mathcal{M}$, every formula φ , and every point (l, θ, o) , the value $\widehat{V}(\varphi, l, \theta, o)$ depends only on the restriction of V to $\text{AP}(\varphi) \times \{l\} \times \{\theta\} \times \{o\}$.*

Proof. Structural induction on φ . The base case ($\varphi = p$) is immediate by the definition of $\widehat{V}$ on atoms. For each compound case $\varphi = \neg\psi$, $\psi \wedge \chi$, $\psi \vee \chi$, $\psi \rightarrow \chi$, the value at (l, θ, o) is a function (the corresponding Kleene connective) of $\widehat{V}(\psi, l, \theta, o)$ and possibly $\widehat{V}(\chi, l, \theta, o)$, which by the induction hypothesis depend only on V restricted as claimed. $\text{AP}(\varphi)$ is the union of the AP of its sub-formulas, so the restriction propagates. $\square$

Proposition 9 (Information-monotonicity). *Let V_1, V_2 be base valuations on the same frame with $V_1(p, l, \theta, o) \sqsubseteq V_2(p, l, \theta, o)$ for every atom and every point. Then for every formula φ and every point (l, θ, o) , $\widehat{V}_1(\varphi, l, \theta, o) \sqsubseteq \widehat{V}_2(\varphi, l, \theta, o)$.*

Proof. Structural induction, using that $\neg_{K_3}, \wedge_{K_3}, \vee_{K_3}, \rightarrow_{K_3}$ are each monotone in $\sqsubseteq$ in each argument (a standard K_3 result, verified by inspection of the truth tables). $\square$

Corollary 10 (Conservativity over classical logic). *Let $\mathcal{M}$ be a model in which V never takes the value $\mathbf{U}$. Then for every formula φ and every point (l, θ, o) , $\widehat{V}(\varphi, l, \theta, o) \in \{\mathbf{T}, \mathbf{F}\}$, and the resulting value coincides with the classical (two-valued) valuation of φ under the same atomic assignment.*

Proof. By Proposition 9 and inspection of the Kleene tables: when neither input is $\mathbf{U}$, every Kleene connective returns the classical value. $\square$

Proposition 11 (De Morgan laws). *For all formulas φ, ψ and all (l, θ, o) ,*

$$\begin{aligned}\widehat{V}(\neg(\varphi \wedge \psi), l, \theta, o) &= \widehat{V}(\neg\varphi \vee \neg\psi, l, \theta, o), \\ \widehat{V}(\neg(\varphi \vee \psi), l, \theta, o) &= \widehat{V}(\neg\varphi \wedge \neg\psi, l, \theta, o).\end{aligned}$$

Proof. By case analysis on $(\widehat{V}(\varphi), \widehat{V}(\psi)) \in \{\mathbf{T}, \mathbf{F}, \mathbf{U}\} \times \{\mathbf{T}, \mathbf{F}, \mathbf{U}\}$ (nine cases per identity). Each case checks directly against the truth tables of Section 2. $\square$

Proposition 12 (Double negation, idempotence, commutativity, associativity). *For all φ, ψ, χ and all (l, θ, o) , $\widehat{V}$ satisfies: $\neg\neg\varphi \equiv \varphi$; $\varphi \wedge \varphi \equiv \varphi$; $\varphi \vee \varphi \equiv \varphi$; $\varphi \wedge \psi \equiv \psi \wedge \varphi$; $\varphi \vee \psi \equiv \psi \vee \varphi$; $(\varphi \wedge \psi) \wedge \chi \equiv \varphi \wedge (\psi \wedge \chi)$; $(\varphi \vee \psi) \vee \chi \equiv \varphi \vee (\psi \vee \chi)$, where $\equiv$ means equality of $\widehat{V}$ -values at every point of every model.*

Proof. Each is a finite truth-table check, lifted point-wise via Lemma 8. $\square$

Theorem 13 (No universal **T**-tautologies). *There is no formula $\varphi \in \mathcal{L}$ such that $\models \varphi$ (i.e., $\widehat{V}(\varphi, l, \theta, o) = \mathbf{T}$ in every model at every point).*

Proof. Consider the model $\mathcal{M}_{\mathbf{U}}$ on any frame with $V(p, l, \theta, o) = \mathbf{U}$ for every atom p and every point. By induction on φ , using the Kleene tables (each connective returns **U** when all inputs are **U**), $\widehat{V}(\varphi, l, \theta, o) = \mathbf{U}$ for every φ and every (l, θ, o) . Hence no formula evaluates to **T** universally. This re-derives, in the indexed setting, the well-known absence of K_3 -tautologies under designated value **T** [2]. $\square$

Corollary 14 (Failure of LEM and of contradiction). $\not\models p \vee \neg p$ and $\not\models \neg(p \wedge \neg p)$.

Proof. Take $\mathcal{M}_{\mathbf{U}}$ as in Theorem 13: both formulas evaluate to **U**, not **T**. $\square$

Remark 15. Theorem 13 is the price of taking **T** as the sole designated value in K_3 . It does not mean the logic is useless—it means valid *inferences* (consequences from non-empty Γ) are the appropriate notion of validity. For example, $\{p, p \rightarrow q\} \models q$ holds: by inspection of the implication and conjunction tables, whenever $\widehat{V}(p) = \widehat{V}(p \rightarrow q) = \mathbf{T}$, we have $\widehat{V}(q) = \mathbf{T}$.

7 Worked Example

Let $L = \{l_{\text{EU}}, l_{\text{US}}\}$, $\Theta = \{\theta_0, \theta_1\}$, $O = \{o_{\text{Alice}}, o_{\text{Bob}}\}$. Let p be the atomic proposition “the package has been delivered” and q “the package has cleared customs.” A possible base valuation:

l	θ	o	$V(p)$	$V(q)$
l_{EU}	θ_0	o_{Alice}	U	F
l_{EU}	θ_1	o_{Alice}	T	T
l_{EU}	θ_1	o_{Bob}	U	T
l_{US}	θ_1	o_{Alice}	F	U

Then at $(l_{\text{EU}}, \theta_1, o_{\text{Alice}})$, $\widehat{V}(p \wedge q) = \mathbf{T}$ (Alice in the EU at θ_1 knows both); at $(l_{\text{EU}}, \theta_1, o_{\text{Bob}})$, $\widehat{V}(p \wedge q) = \mathbf{U}$ (Bob does not yet know whether the package has been delivered, even though he knows it has cleared customs); at $(l_{\text{US}}, \theta_1, o_{\text{Alice}})$, $\widehat{V}(p \wedge q) = \mathbf{F}$ (the proposition is false in the US frame because $\widehat{V}(p) = \mathbf{F}$). Different points yield different truth values for *the same formula*—this is the design.

8 Related Work

The components of LTO- K_3 are each well-studied.

Three-valued logics. K_3 [2] is one of several three-valued logics with different motivations and different connectives. Łukasiewicz $\mathbb{L}_3$ [1] differs from K_3 in implication: $\mathbf{U} \rightarrow_{\mathbb{L}_3} \mathbf{U} = \mathbf{T}$, which yields the tautology $p \rightarrow p$ and other classical theorems. Bochvar’s internal B_3 [9] is “infectious”: any operation with a **U** input returns **U**, even $\mathbf{T} \vee \mathbf{U}$. Priest’s LP [8] uses the Kleene tables but designates $\{\mathbf{T}, \mathbf{U}\}$, yielding a paraconsistent logic in which contradictions can be true and LEM holds. Belnap’s **FOUR** [6] adds a fourth value (“both true and false”) and is appropriate when contradictory information is to be tolerated rather than collapsed to **U**.

We choose K_3 with designated value **T** because the reading of **U** we want is purely epistemic (“not yet known”) and information-monotonic (Proposition 9): resolving an **U** to a classical value never overturns a previously determined truth. L_3 breaks this property (its implication is not monotone in the obvious sense); LP changes the designated set; B_3 propagates **U** too aggressively for our use case; **FOUR** conflates uncertainty with inconsistency.

Temporal and modal logics. Prior [3] introduced tense operators; Pnueli [7] introduced LTL for program verification; Clarke and Emerson [10] introduced CTL. These logics index truth by time (or by states reachable in time) but use classical two-valued semantics and quantify over states with modal operators. Kripke [5] gave possible-worlds semantics for modal logic; the entire modal-logic tradition [12] indexes truth by worlds and uses $\Box, \Diamond$ to quantify over them. Hybrid logic [14] adds nominals i and the operator $@_i$ to refer to specific worlds in the object language. LTO-K3 differs by *not* quantifying over points in the object language at all—there are no $\Box, \Diamond, F, G, K_o, @_i$. Indices live only in the meta-language.

Epistemic logics. Hintikka [4] introduced epistemic modal logic with K_a operators. The textbook treatment is Fagin et al. [11]. Multi-agent epistemic temporal logic combines these dimensions, but again with two-valued semantics and modal operators. Three-valued epistemic logics exist but are typically formulated with modal operators rather than as a flat indexed valuation.

Three-valued spatio-temporal and runtime-verification logics. Bauer et al. [15] developed LTL₃, a three-valued LTL variant for runtime monitoring with truth values true, false, and inconclusive. Spatial logics [13] introduce explicit spatial indices for reasoning about topological structure. These works are the closest neighbours of LTO-K3.

Honest positioning. Relative to the above, LTO-K3 is best read as a deliberately impoverished system. It has fewer expressive resources than hybrid logic, fewer than multi-modal epistemic temporal logic, and fewer than LTL₃. What it offers in return is a uniform treatment of three indices (location, time, observer) as concrete and equally first-class, and a self-contained semantics that fits on a page. We make no claim that LTO-K3 supersedes any of these systems; we claim only that it is a useful *starting point* when their full machinery is not required.

9 Limitations

1. **No cross-frame operators.** LTO-K3 cannot express “ φ at l_1 implies ψ at l_2 ” or “ φ now will be ψ later.” Such statements require modal operators, hybrid nominals, or explicit bridging rules, none of which are part of the core.
2. **No quantification.** LTO-K3 cannot express “ φ holds for some observer” or “ φ holds at all times.” These require either propositional quantifiers or meta-quantification.
3. **Structure on L, Θ, O is invisible.** Time is not ordered, locations are not topologized, observers do not have information-state relations. Any such structure must be added by an extension and cannot be queried within $\mathcal{L}$.
4. **No T-tautologies** (Theorem 13). Validity must be reasoned about via non-trivial entailments, not via tautologies. Users accustomed to classical-logic tautological reasoning will need to adjust.
5. **U is undifferentiated.** “Not yet known”, “in principle unknowable”, and “not yet defined” are collapsed into a single value. Distinguishing them requires a richer truth-value set (e.g., [6]) or a probabilistic / possibilistic layer.

6. **No interaction between observers.** There is no notion of “ o_1 knows that o_2 knows φ ”; that requires an explicit K_o operator and is intentionally outside scope.
7. **The cross-frame undefinedness is a restriction, not a feature.** We have framed it as a deliberate design choice, and it is, but it is also a real expressive limitation. Users who need cross-frame reasoning will need to layer on additional machinery.

10 Conclusion

We have presented LTO-K3, a minimal three-valued logic in which every proposition is evaluated under an explicit triple of location, time, and observer. The system is a careful packaging of standard ingredients—Kleene’s K_3 , indexed valuations, designated-value consequence—into a single self-contained core, together with a small body of meta-theoretic properties (locality, monotonicity, conservativity, De Morgan, absence of **T**-tautologies). We have been explicit about what is new (the packaging) and what is not (everything else), and have named the system’s limitations. Cross-frame composition, quantification, and modal operators are deliberate omissions to be addressed by domain-specific extensions.

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